

LOUISVILLE INTL, KY, USA

WMO: 724230

Lat: 38.181N

Lon: 85.739W

Elev: 488

StdP: 14.44

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93821

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.2	15.7	-2.2	5.0	13.5	2.9	6.5	18.3	25.6	42.1	23.3	42.2	8.3	300	0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	94.0	75.1	91.7	74.8	89.6	74.0	78.6	89.0	77.4	87.7	76.3	85.9	9.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.6	136.4	84.9	74.3	130.4	83.4	73.2	125.5	82.1	42.4	89.0	41.0	87.5	39.9	86.2	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.5	18.8	16.9	3.9	97.3	7.5	3.6	-1.5	99.9	-5.9	102.0	-10.1	104.0	-15.5	106.6		
(5)			2.6	80.7	7.1	1.5	-2.5	81.8	-6.7	82.7	-10.7	83.6	-15.8	84.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.6	35.0	39.1	47.9	58.9	67.9	76.1	79.2	78.6	71.8	60.1	48.2	39.4
	DBStd	17.91	12.03	11.33	10.69	9.20	7.73	5.54	4.68	4.68	7.11	8.72	9.46	10.47
	HDD50	1535	487	335	170	26	1	0	0	0	0	15	144	357
	HDD65	4010	931	726	538	224	63	3	0	0	21	203	507	794
	CDD50	4678	21	30	105	292	556	782	904	887	654	328	91	28
	CDD65	1678	0	1	8	41	153	335	439	422	225	50	3	1
	CDH74	16045	0	4	60	333	1251	3166	4550	4194	2058	411	18	0
	CDH80	6153	0	0	5	57	359	1213	1911	1726	781	100	1	0
Wind	WSAvg	7.7	8.9	8.9	8.9	9.1	7.7	7.1	6.4	6.1	6.1	6.9	7.8	8.1
Precipitation	PrecAvg	46.0	3.1	3.3	4.6	4.4	4.8	3.9	4.3	3.5	3.5	3.1	3.6	3.9
	PrecMax	69.0	7.2	10.6	14.9	14.0	11.6	10.1	10.0	8.8	10.9	10.0	7.6	8.9
	PrecMin	32.6	0.4	0.7	1.0	0.7	1.4	0.5	0.4	0.6	0.1	0.4	0.7	0.6
	PrecStd	8.1	1.6	2.0	2.6	2.6	2.3	2.1	2.1	1.9	2.4	2.1	1.7	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.4	71.9	80.1	85.0	89.3	93.6	97.3	97.2	95.7	87.3	76.4	69.4
		MCWB	60.8	60.5	63.2	65.4	72.3	75.4	77.4	75.0	71.4	69.8	63.5	61.6
	2%	DB	62.9	66.3	74.9	81.5	86.5	91.3	93.7	93.6	91.4	82.5	71.7	64.0
		MCWB	57.6	57.0	60.2	64.3	70.2	75.0	76.6	74.7	71.7	67.7	60.7	58.0
	5%	DB	58.7	62.0	69.9	78.0	84.3	89.2	91.4	91.0	88.1	78.7	67.5	59.4
		MCWB	54.1	54.4	57.7	62.3	69.3	73.8	75.9	74.6	71.1	65.4	58.6	54.0
	10%	DB	53.6	57.0	65.3	74.4	81.4	87.0	89.0	88.4	84.5	74.6	63.6	55.3
		MCWB	48.3	49.5	55.0	61.0	68.5	72.8	74.8	73.5	69.6	63.5	56.0	50.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.3	62.9	64.9	69.9	75.0	78.9	81.3	80.3	76.2	73.6	66.3	63.0
		MCDB	66.1	69.9	75.7	80.4	84.8	89.1	91.2	90.5	87.1	84.1	72.7	68.1
	2%	WB	58.9	59.5	62.2	67.3	73.1	77.4	79.1	78.2	74.7	70.8	63.4	59.1
		MCDB	62.3	64.9	71.5	76.8	83.0	88.0	90.0	88.3	85.3	78.4	69.2	63.4
	5%	WB	54.8	55.3	59.8	65.1	71.5	76.1	77.8	76.9	73.3	68.0	60.1	55.1
		MCDB	58.2	60.8	67.8	73.5	81.2	86.0	88.3	86.7	83.0	75.8	66.0	58.7
	10%	WB	49.3	50.1	56.6	62.9	69.9	74.8	76.6	75.5	72.0	65.3	56.6	50.8
		MCDB	53.6	56.0	64.3	71.5	78.6	84.1	86.2	84.6	80.9	72.7	62.4	54.5
Mean Daily Temperature Range	5% DB	MDBR	14.9	16.3	18.2	19.8	17.9	17.0	16.8	17.5	19.1	19.0	17.1	14.1
		MCDBR	19.0	22.1	24.1	23.3	20.2	19.1	19.0	20.2	22.4	22.7	21.9	18.8
		MCWBR	15.3	16.2	13.7	11.7	8.7	7.7	6.8	7.1	7.9	10.7	14.1	15.0
	5% WB	MCDBR	17.6	20.2	21.0	19.5	17.4	17.1	17.0	16.8	17.8	18.4	18.8	16.9
MCWBR		15.2	16.8	13.8	11.9	8.8	7.9	7.5	7.1	7.8	10.2	14.5	15.1	
Clear-Sky Solar Irradiance	taub	0.315	0.330	0.360	0.405	0.445	0.458	0.482	0.479	0.423	0.364	0.331	0.309	
	taud	2.458	2.390	2.380	2.255	2.184	2.184	2.126	2.122	2.255	2.425	2.463	2.482	
	Ebn at Noon	274	284	285	277	267	263	255	252	261	268	266	268	
	Edh at Noon	26	32	35	43	46	46	49	48	40	30	26	24	
All-Sky Solar Radiation	RadAvg	609	830	1175	1558	1738	1974	1874	1750	1442	1072	733	523	
	RadStd	64	115	110	109	156	112	114	101	134	115	81	52	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.95	N/A	N/A	N/A	N/A	-0.68	N/A	N/A	+281	+202
(47) Regional (30 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+212	+139

Nomenclature: See separate page